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Batch: A1 RollNo: 18

```
import time
import pandas as pd
from sklearn.datasets import fetch_california_housing
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
```

```
data = fetch_california_housing()
print(data.keys())
```

```
dict_keys(['data', 'target', 'frame', 'target_names', 'feature_names', 'DESCR'])
```

```
x = pd.DataFrame(data.data, columns = data.feature_names)
y = data.target
print(x.head())
print("samples:", len(x))
print(y[:10])
```

	MedInc	HouseAge	AveRooms	...	AveOccup	Latitude	Longitude
0	8.3252	41.0	6.984127	...	2.555556	37.88	-122.23
1	8.3014	21.0	6.238137	...	2.109842	37.86	-122.22
2	7.2574	52.0	8.288136	...	2.802260	37.85	-122.24
3	5.6431	52.0	5.817352	...	2.547945	37.85	-122.25
4	3.8462	52.0	6.281853	...	2.181467	37.85	-122.25

```
[5 rows x 8 columns]
samples: 20640
[4.526 3.585 3.521 3.413 3.422 2.697 2.992 2.414 2.267 2.611]
```

```
x_train,x_test,y_train,y_test = train_test_split(x, y, test_size = 0.2, random_state= 42)
```

```
tree = DecisionTreeRegressor()
start = time.perf_counter()
tree.fit(x_train, y_train)
tree_time = time.perf_counter() - start
pred_tree_test = tree.predict(x_test)
err_tree_test = mean_absolute_error(y_test, pred_tree_test)
pred_tree_train = tree.predict(x_train)
err_tree_train = mean_absolute_error(y_train, pred_tree_train)
print("Decision Tree Train MAE:" , err_tree_train)
print("Decision Tree Test MAE:", err_tree_test)
print("Decision Tree Time:", tree_time)
```

```
Decision Tree Train MAE: 4.217126217568424e-17
Decision Tree Test MAE: 0.45369783430232563
Decision Tree Time: 0.3055197629998929
```

```
rf = RandomForestRegressor()
start = time.perf_counter()
rf.fit(x_train, y_train)
```

```

rf_time = time.perf_counter() - start
pred_rf_test = rf.predict(x_test)
err_rf_test = mean_absolute_error(y_test, pred_rf_test)
pred_rf_train = rf.predict(x_train)
err_rf_train = mean_absolute_error(y_train, pred_rf_train)
print("Random Forest Train MAE:" , err_rf_train)
print("Random Forest Test MAE:", err_rf_test)
print("Random Forest Time:", rf_time)

```

```

Random Forest Train MAE: 0.1221313927143899
Random Forest Test MAE: 0.3262400572189924
Random Forest Time: 21.99320275199989

```

```

gb = GradientBoostingRegressor(n_estimators = 300, learning_rate = 0.1, max_depth = 5)
start = time.perf_counter()
gb.fit(x_train, y_train)
gb_time = time.perf_counter() - start
pred_gb_test = gb.predict(x_test)
err_gb_test = mean_absolute_error(y_test, pred_gb_test)
pred_gb_train = tree.predict(x_train)
err_gb_train = mean_absolute_error(y_train, pred_gb_train)
print("Gradient Boosting Train MAE:" , err_gb_train)
print("Gradient Boosting Test MAE:", err_gb_test)
print("Gradient Boosting Time:", gb_time)

```

```

Gradient Boosting Train MAE: 4.217126217568424e-17
Gradient Boosting Test MAE: 0.3065082611923547
Gradient Boosting Time: 49.57473926600005

```

```

print("\n-----Comparison For Test MAE-----")
print("Decision Tree Test:", err_tree_test)
print("Random Forest Test:", err_rf_test)
print("Gradient Boosting Test:", err_gb_test)

```

```

-----Comparison For Test MAE-----
Decision Tree Test: 0.45369783430232563
Random Forest Test: 0.3262400572189924
Gradient Boosting Test: 0.3065082611923547

```

```

print("\n-----Comparison For Train MAE-----")
print("Decision Tree Train:", err_tree_train)
print("Random Forest Train:", err_rf_train)
print("Gradient Boosting Train:", err_gb_train)

```

```

-----Comparison For Train MAE-----
Decision Tree Train: 4.217126217568424e-17
Random Forest Train: 0.1221313927143899
Gradient Boosting Train: 4.217126217568424e-17

```

```

print("\n-----Comparison For Time MAE-----")
print("Decision Tree time:", tree_time)
print("Random Forest time:", rf_time)
print("Gradient Boosting Time:", gb_time)

```

```

-----Comparison For Time MAE-----
Decision Tree time: 0.3055197629998929
Random Forest time: 21.99320275199989
Gradient Boosting Time: 49.57473926600005

```